



TAA System

Gestión de Portafolios

Mayo 2026

Agenda

- 1 System Overview**
- 2 Signal Engine & Transform Codes**
- 3 The Four Pillars (F / M / S / V)**
- 4 Composite Scoring & Conviction Mapping**
- 5 Conviction & Views**
- 6 Hierarchy, Portfolio & Implementation**

Design Principles and Active Asset Class Universe

The TAA system generates weekly tactical tilts across 6 active asset classes using 4 signal pillars

Design Principles

- **Inputs** — All signals, weights, and mappings defined in config/taa_config.xlsx. No hardcoded parameters in code.
- **Signals** — Every signal has a verifiable data source (Bloomberg, FRED, ICE BofA) and clear economic rationale. There could be custom signals that combine plain inputs from approved sources, allowing analytical flexibility without losing transparency.
- **Sign lives in mapping** — Direction (bullish/bearish) is set exclusively in SignalMapping.
- **No short positions** — force_zero_sum = True, weights clipped at zero. MM and STFI absorb net tilts.
- **Full reproducibility** — All outputs time-stamped (RUN_YYYYMMDD_HHMMSS). Inputs + config version-controlled. Pipeline produces identical outputs given identical inputs.

Active Asset Class Universe

Asset Class	Group	Type
LT US Treasuries	FI	L2 child of FI agg
LT US Corporate	FI	L2 child of FI agg
LT EM Fixed Income	FI	L2 child of FI agg
US Equity	EQ	Standalone L1
DM ex-US Equity	EQ	Standalone L1
EM Equity	EQ	Standalone L1

Note: Money Market and Short-Term FI are present in portfolios with SAA weights but receive **no TAA signal tilt**. They naturally absorb the net excess of tilts generated by the 6 active ACs under the zero-sum constraint.

For example, an OW in EM Equity is effectively financed by a relative UW in MM/STFI.

All 6 ACs roll up into 4 L1 buckets for the hierarchy view.

Construction of the TAA System

From signal pillars to Z-score aggregation and final absolute/relative views

Methodological architecture of the TAA engine

Weekly signal system for active tactical tilts around strategic allocations

#	Stage	Input	Output	Main logic
1	Signal pillars	Macro, price, sentiment and valuation series	F / M / S / V scores	Signals grouped by economic role
2	Signal transforms	Raw time series	Comparable z-scores	EWMA, rolling, percentile and momentum transforms
3	Pillar construction	Signed signal z-scores	Pillar z-score	Weighted average + re-standardisation
4	Composite scoring	F/M/S/V pillar scores	Composite z-score	AC-specific pillar weights
5	Views	Composite z-score	Absolute and relative tilts	35/65 blend of absolute and cross-sectional views

System scope

6

Active asset classes
3 FI + 3 Equity sleeves

4

Signal pillars
Fundamentals, Momentum, Sentiment, Valuation

W

Weekly engine
Outputs scorecard, composites and portfolio views

±

Sign in mapping
Direction is assigned by asset class and pillar

Σ

Z-score framework
All signals become comparable before aggregation

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Signal Transform Framework: From Raw Data to Comparable Z-Scores

Six transform codes convert raw series into comparable z-scores with uniform sign conventions

Signal Flow Pipeline

taa_config.xlsx (170 DataSeries rows) → load raw → apply transform_code → clip to ≥ 2013-02-01 → clip outliers to ±3σ → {series_id: z-score Series}

Transform Codes

Code	Output	Formula	Use Cases
ewma_z	EWMA z-score	$(x - \text{EWMA_mean}) / \text{EWMA_std}$, span = 756d	VIX, DXY, FCI, PMI, sentiment
rolling_z	Rolling z-score	$(x - \text{roll_mean}) / \text{roll_std}$, window per row	Breakevens, ERP, term spread
pctile	Percentile rank	$\text{rank}(x)/n \rightarrow (p - 0.5) \times 4$	OAS levels, yield levels
mom_z	Momentum z-score	$\text{ewma_z}(\text{pct_change}(\text{window}))$	Forward EPS revisions
price_mom	Composite momentum	$40\% \times 12\text{-}1\text{M} + 25\% \times 3\text{M} + 25\% \times \text{MA} + 10\% \times \text{RSI}$	TR price indices (EQ and FI)
inv_mom_z	Inverse momentum z	$-\text{ewma_z}(\text{diff}(\text{window}))$, falling = positive	OAS spreads, Treasury yields

Sign Convention

+1 = series value ↑ → bullish for this AC in this pillar | -1 = series value ↓ → bearish. Signs set exclusively in SignalMapping. Raw series never inverted in Python.
Example: PMI US is +1 for US Equity but -1 for LT Treasuries.

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Fundamentals & Momentum

Pillar F captures macro regime signals while Pillar M tracks price and spread trends

Pillar F — Fundamentals (Macro Engine)

Series	Transform	Direction Logic
PMI US / EZ / China composite	ewma_z	Expansion = bullish EQ/credit
CESI (Surprise indices)	ewma_z	Positive surprise = bullish EQ
GDP blend (consensus)	ewma_z	Growth upgrade = bullish
GDPNow (real-time)	ewma_z	US short-term growth proxy
Forward EPS growth/revision	mom_z, ewma_z	EPS acceleration = bullish
ISM N.O. / Inventories	ewma_z	>1.0 = demand > inventory
Breakeven inflation (1Y/5Y/10Y)	rolling_z	Higher BEI = bearish duration
Real Fed Funds Rate	rolling_z	Restrictive = bearish
Core PCE YoY	ewma_z	Above target = Fed restrictive

All Fundamentals signals use ewma_z or rolling_z. These are slow-moving regime indicators designed to capture the macroeconomic environment.

Pillar M — Momentum (Price/Spread Trends)

Series	Transform	Direction Logic
TR price indices (SP500, EAFE, EM)	price_mom	Composite multi-factor
FI TR indices (BSGV, BFU5, LT03)	price_mom	Falling yield = rising price
OAS momentum (BBB, HY, EM)	inv_mom_z	Tightening = positive
Treasury yield momentum (2Y, 10Y)	inv_mom_z	Yield falling = positive
CDX IG / HY momentum	ewma_z	CDS tightening = positive
EPS revision momentum	mom_z	40%×3M + 60%×6M blend

Momentum captures price and spread trend persistence. The composite price_mom blends 12-1M skip (40%), 3M (25%), MA cross (25%), and RSI (10%).

Pillar Aggregation Formula

Step 1 — Weighted composite: $composite_t = \sum_i (sign_i \times weight_i \times raw_z_i(t)) / \sum_i weight_i$

Step 2 — Re-standardise: $pillar_z_t = (composite_t - EWMA_mu) / EWMA_std, span=756$

Re-standardising ensures unit variance before composite combination. The Pillar Z-Score is the EWMA z-score of the weighted sum — not a simple weighted average of signal z-scores.

Sentiment and Valuation

Pillar S measures risk appetite and positioning while Pillar V assesses absolute and relative cheapness

Pillar S — Sentiment (Risk Appetite)

Series	Transform	Direction Logic
CBOE VIX	ewma_z	High = contrarian buy EQ
MOVE Index (bond vol)	ewma_z	High = flight to UST
VSTOXX	ewma_z	EZ-specific fear
SKEW Index	ewma_z	Tail risk indicator
Modern TED (T-bill – SOFR)	ewma_z	Funding stress; gated 2018+
OAS stress proxies (diff 21d)	ewma_z	HY/EM widening = risk-off
AAll Bull-Bear spread	ewma_z	Contrarian: extreme bull = sell
CBOE Put/Call ratio	ewma_z	High PCR = fear = buy
Bloomberg US/EZ FCI	ewma_z	Tight = headwind for EQ/credit
DXY USD Index	ewma_z	Strong = EM headwind

Pillar V — Valuation

Series	Transform	Direction Logic
PE scores (SP500, EAFE, EM)	ewma_z	Low P/E = cheap = bullish
Equity Risk Premium	rolling_z	High ERP = equity cheap vs bonds
Relative PE (Gro/Val, DM/US)	ewma_z	Relative cheapness signal
OAS levels (BBB, HY, EM, LatAm)	pctile	High pctile = cheap credit
HY/IG ratio	ewma_z	High ratio = HY expensive
Yield levels (2Y, 10Y, TIPS)	pctile	High yield = attractive carry
Term spread (10Y – 2Y)	ewma_z	Inverted = recession = bullish UST
Real Fed Funds Rate	rolling_z	Restrictive policy = bearish

Pillar Weights by Asset Classes

~ Balanced across pillars (25% each one)

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Composite Z-Score Aggregation & Conviction Scaling

Pillar z-scores are aggregated, re-standardized, and combined into composite scores with conviction multipliers

COMPOSITE Z-SCORE FORMULA

$Z_{\text{composite}} = \sum_p (w_p \times Z_{\text{pillar}_p}) / \sum_p w_p$

Missing pillars skipped gracefully – weights renormalize to available pillars.
Final composite clipped to $\pm 3\sigma$.

Pillar weights are AC-specific and sum to 1.0 per row, stored in PillarWeights sheet.

CONVICTION MULTIPLIER

Pillars in Agreement	Multiplier	Conviction
4 / 4	1.00×	Full
3 / 4	0.80×	High
2 / 4	0.50×	Moderate
1 / 4 or 0	0.00×	None

A pillar "agrees" if its z-score has the same sign as the composite AND $|z_{\text{pillar}}| > 0.25$.
With 1/4 agreement, no tilt is applied even if composite is large.

WHY RE-STANDARDISE? – ENSURING UNIT VARIANCE

- The weighted average of individually z-scored signals can still have non-unit variance due to correlations between signals and partial data availability. Re-standardising via `standardise_pillar()` ensures every pillar has unit variance before being combined into the composite.
- This means the Pillar Z-Score displayed in the dashboard is not a simple weighted sum of the signal z-scores – it is the historical z-score (EWMA-standardised) of that weighted sum against its own history. This critical step ensures comparability across pillars and prevents any single pillar from dominating due to variance differences.

CRISIS OVERRIDE

When both VIX and MOVE exceed 80th percentile simultaneously, all tilts forced to zero.

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From composite z-score to tactical intensity

Conviction thresholds map composite z-scores to discrete tilt levels with crisis override protection

CONVICTION THRESHOLDS

Z range	Label	Tilt fraction	Color	Pillars in agreement	Multiplier
$z \geq +1.50$	HIGH OW	+1.0	● Strong Bull	4 / 4	1.00
+0.75 to +1.49	MEDIUM OW	+0.5	● Moderate Bull	3 / 4	0.80
-0.75 to +0.74	NEUTRAL	0.0	● Neutral	2 / 4	0.50
-1.50 to -0.76	MEDIUM UW	-0.5	● Moderate Bear	1 / 4	0.00
$z < -1.50$	HIGH UW	-1.0	● Strong Bear	0 / 4	0.00

ABSOLUTE TILT FORMULA

$$\text{abs_tilt} = \text{tilt_fraction} \times \text{conviction_multiplier} \times \text{MAX_TILT_PCT[AC]}$$

Where:

- **tilt_fraction** = from threshold table (+1.0, +0.5, 0.0, -0.5, -1.0)
- **conviction_multiplier** = pillar agreement (1.00×, 0.80×, 0.50×, 0.00×)
- **MAX_TILT_PCT[AC]** = max tilt at 100 bps TE reference

MAX_TILT_PCT is auto-generated from taa_config.xlsx and varies by asset class based on risk budget allocation.

CRISIS OVERRIDE MECHANISM

When both **VIX > 80th percentile** AND **MOVE > 80th percentile** simultaneously → all tilts forced to zero. Override lifts when both return below 70th percentile. This protects against environments where cross-asset correlation spikes and diversification breaks down – the exact scenario where systematic signals are most vulnerable.

Thresholds calibrated for insurance portfolio context with Solvency II constraints. No-short mandate enforced via clip(lower=0).

Absolute vs Relative Views & Final Tilt Blend

Final tilts blend 35% absolute views with 65% relative cross-sectional rankings

Final view framework

Absolute view

Answers: “Is this asset class attractive relative to its own history?” It uses the composite z-score directly and converts it into an absolute tilt.

$$Z_{\text{absolute}} = Z_{\text{composite}}$$

$$\text{abs_tilt} = f(Z_{\text{absolute}}) \times \text{conviction_mult} \times \text{MAX_TILT}$$

Uses the AC's own composite z-score against its historical distribution. Captures time-series mean reversion and trend persistence within each asset class independently.

Relative view

Answers: “Which asset class looks better than the others today?” It rescales each composite score relative to the cross-sectional mean and dispersion.

$$Z_{\text{relative}} = (Z_{\text{composite}} - \text{mean_cs}) / \text{std_cs}$$

$$\text{rel_tilt} = f(Z_{\text{relative}}) \times \text{conviction_mult} \times \text{MAX_TILT}$$

Cross-sectional ranking across all 6 active ACs. mean_cs and std_cs computed cross-sectionally. Captures which assets are cheap vs peers.

FINAL TILT BLEND

$$\text{final_tilt} = 0.35 \times \text{abs_tilt} + 0.65 \times \text{rel_tilt}$$

The 65% weight on relative views tries to capture tactical deviations driven by cross-asset attractiveness rather than absolute macro timing.

KEY ACADEMIC REFERENCES

Wang & Kochard (2012) — 35/65 blend | Asness et al. (2013) — value + momentum | Koijen et al. (2018) — carry is universal